Summarizing Performance: Mean, Median and Mode

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Based on slides created by Mark Hill and others

Summarizing Performance

- Indices of central tendency: Tries to capture “center” of a distribution of values
- Use this “center” to summarize overall behavior. Example
  - Sample mean: Common “average”
  - Median: ½ of the values are above, ½ below
  - Mode: Most common value
- Other means
  - Arithmetic
  - Harmonic
  - Geometric

Motivation and Challenges

- Why mean? We desire to reduce performance to a single number
  - Makes comparisons easy
  - People like a measure of "typical" performance

- Systems are often specialized
  - Performs great on application type X
  - Performs lousy on anything else
- Potentially a wide range of execution times on one system using different benchmark programs
- *How to (correctly) summarize a wide range of measurements with a single value?*

Sample mean

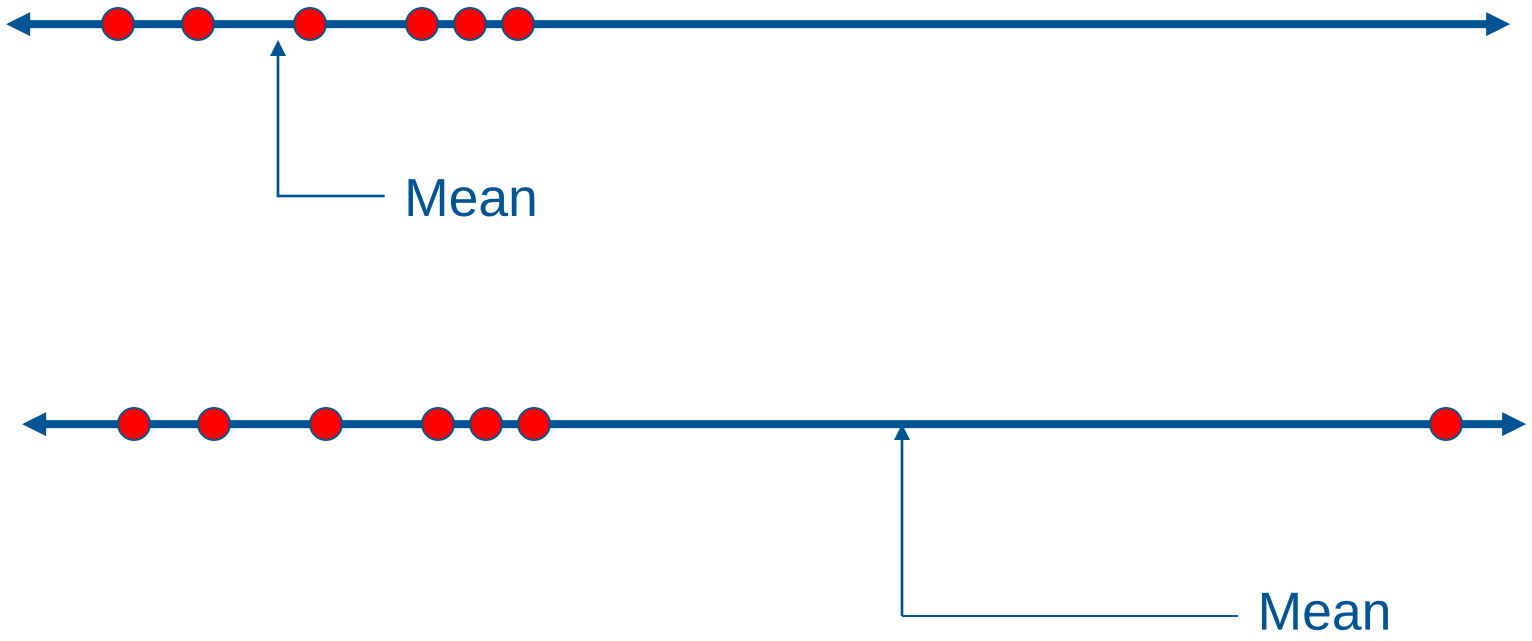
- assume
  - n = number of measurements
- ***Arithmetic mean***
  - Common “average”

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Potential Problem with Means

- Sample mean gives equal weight to all measurements
- *Outliers* can have a large influence on the computed mean value
- Distorts our intuition about the *central tendency* of the measured values

Potential Problem with Means



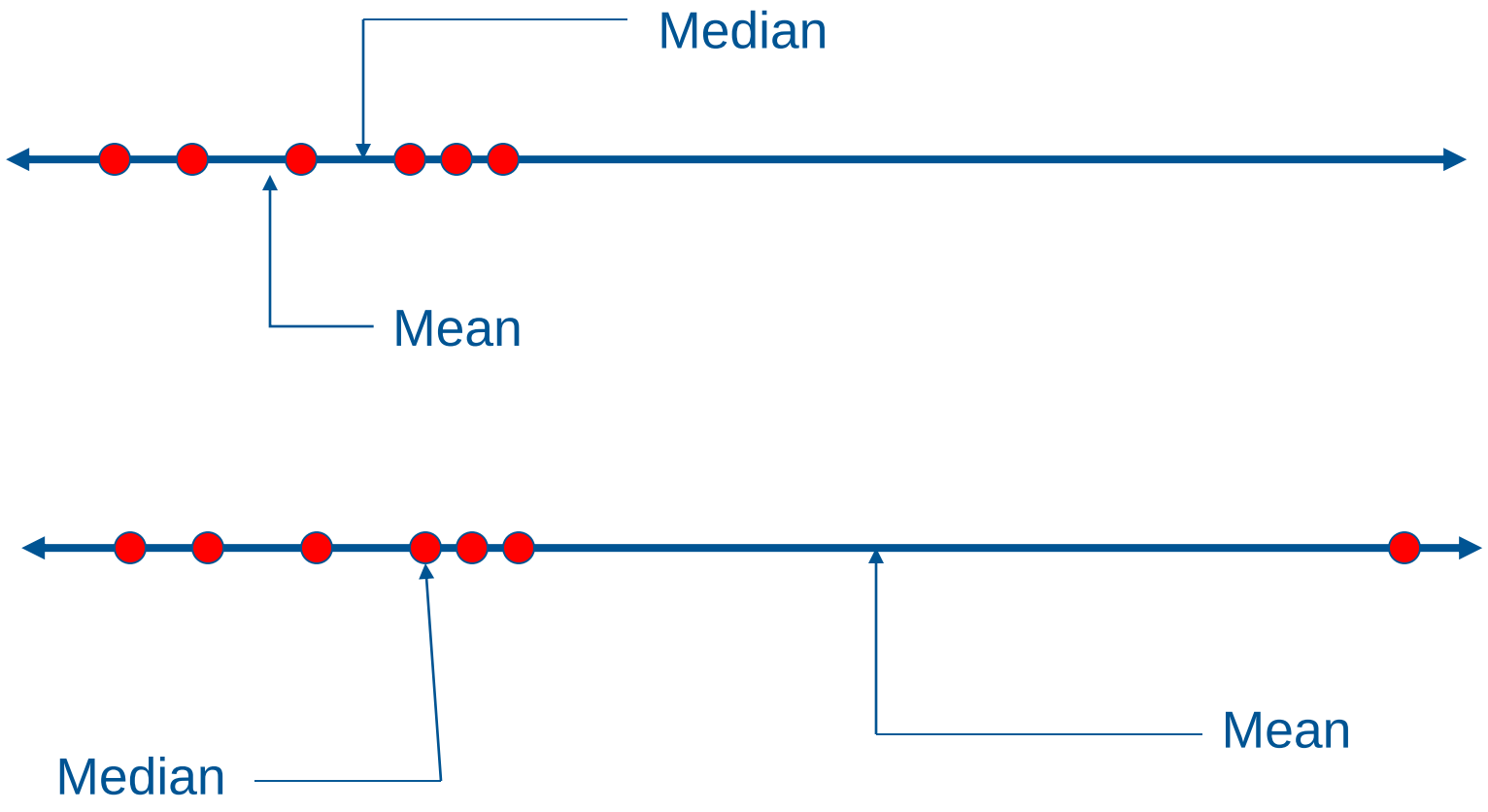
Median

- Index of central tendency with
  - ½ of the values larger, ½ smaller
- Sort n measurements
- If n is odd
  - Median = middle value
  - Else, median = mean of two middle values
- *Reduces skewing effect of outliers* on the value of the index

Example

- Measured values: 10, 20, 15, 18, 16
  - Mean = 15.8
  - Median = 16
- Obtain one more measurement: 200
  - Mean = 46.5
  - Median = ½ (16 + 18) = 17
- Median give more intuitive sense of central tendency

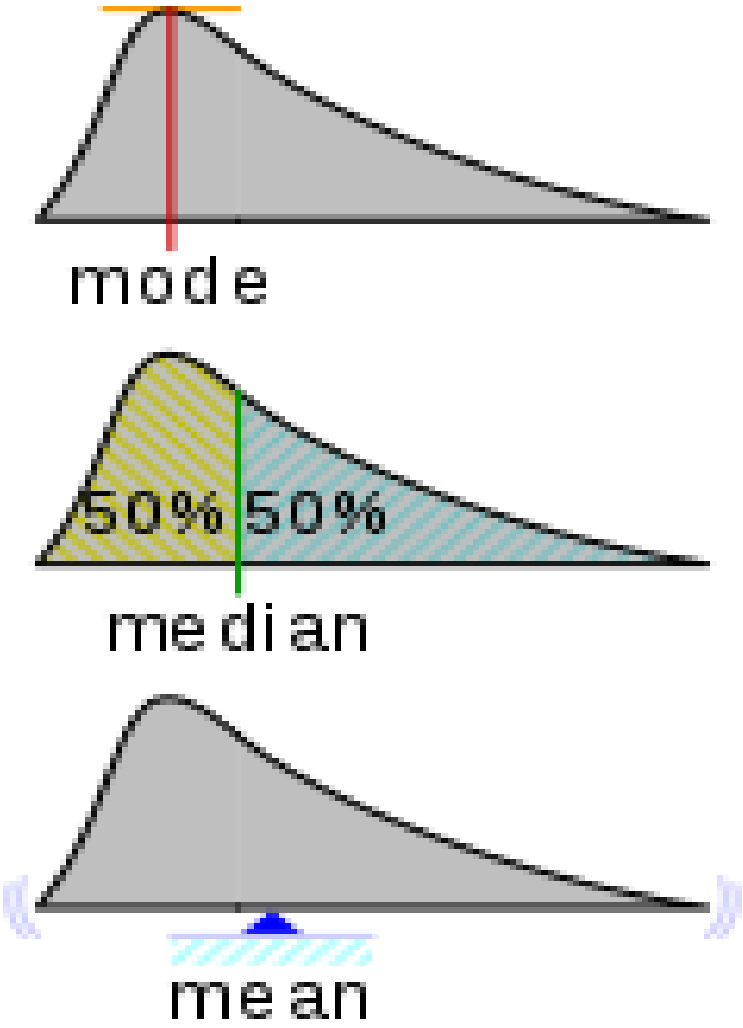
Potential Problem with Means



Mode

- Value that occurs most often
- May not exist
- May not be unique
  - E.g. “bi-modal” distribution
    - Two values occur with same frequency

Mean, Median, or Mode?

- Mean
  - If the sum of all values is meaningful
  - Incorporates all available information
- Median
  - Intuitive sense of central tendency with outliers
  - What is “typical” of a set of values?
- Mode
  - When data can be grouped into distinct types, categories (*categorical data*)

Visualization